

2-D Channel Flow with Artificial Compressibility

ME 7751 Homework 4

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Scripts Included

`channel_flow.py`, `plot_csv.py`, `part3.sh`, `part4.sh`, `part6.sh`, `part7.sh`.

Usage Instructions

All Bash scripts call functionality to `channel_flow.py` and `plot_csv.py`. Run any script to regenerate all the images used for its corresponding part. Run `python channel_flow.py -h` to see available command-line arguments.

Problem 1

Problem Statement

Consider 2-D steady-state channel flow of viscous, incompressible Newtonian fluid with density ρ and dynamic viscosity μ . The channel has height H and length L , and the uniform inlet velocity is U . The flow is governed by the dimensionless continuity and Navier-Stokes equations.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (1)$$

$$\frac{\partial u}{\partial \tilde{t}} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{Re} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad (2)$$

$$\frac{\partial v}{\partial \tilde{t}} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{Re} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) \quad (3)$$

The domain is $(x, y) \in [0, L] \times [0, H]$. By setting the length scale $H = 1$, the time scale $H/U = 1$ (or $U = 1$), and the mass scale $\rho H^3 = 1$ (or $\rho = 1$), we effectively non-dimensionalized all given quantities. Additionally, the Reynold number $Re = \rho U H / \mu = U H / \nu$ is completely determined by the kinematic viscosity $\nu = \mu / \rho$.

At the inlet $x = 0$, $\frac{\partial p}{\partial x} = 0$, $u = U$, $v = 0$. At the outlet $x = L$, $p = 0$, $\frac{\partial u}{\partial x} = 0$, $\frac{\partial v}{\partial x} = 0$. No-slip boundary conditions are set at the walls $y = 0$, $y = H$. In other words, $\frac{\partial p}{\partial y} = 0$, $u = 0$, $v = 0$.

1. Description of Artificial Compressibility Method

Artificial compressibility is a numerical technique used to solve incompressible flow problems by introducing a fictitious time derivative of pressure. This allows the use of compressible flow solvers to find the steady-state solution of incompressible flows without directly solving the pressure Poisson equation, which can be computationally expensive. We begin with the dimensionless 2-D continuity equation for incompressible flow:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (4)$$

By introducing an artificial compressibility term to (4), we make (4) look like the continuity equation for compressible flow:

$$\frac{\partial \tilde{\rho}}{\partial \tilde{t}} + \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (5)$$

$$\frac{\partial p}{\partial \tilde{t}} + \frac{1}{\beta} \frac{\partial u}{\partial x} + \frac{1}{\beta} \frac{\partial v}{\partial y} = 0 \quad (6)$$

Here, $\tilde{\rho}^{-1}$ is the artificial compressibility and $\tilde{t}$ is the artificial time. By relating $\tilde{\rho}$ and the pressure field p according to the equation of state $p = \frac{\tilde{\rho}}{\beta}$ (where $\beta = 0.25$ is a user-defined compressibility constant), we can solve for the pressure field iteratively until the solution converges to a steady state, avoiding the need to solve the Poisson pressure equation directly. As the solution converges, the artificial time derivative of pressure $\frac{\partial p}{\partial \tilde{t}}$ approaches zero, and we recover the incompressible flow solution.

Now, (6) can be used in conjunction with the Navier-Stokes equations. Given a high Reynolds number $Re \gg 1$, the dimensionless Navier-Stokes equations along the x - and y -directions are:

$$\frac{\partial u}{\partial \tilde{t}} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{Re} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad (7)$$

$$\frac{\partial v}{\partial \tilde{t}} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{Re} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) \quad (8)$$

By rearranging, the governing equations (6)-(8) can be rewritten into a more compact form:

$$\frac{\partial \mathbf{U}}{\partial \tilde{t}} + \frac{\partial \mathbf{E}}{\partial x} + \frac{\partial \mathbf{F}}{\partial y} = \frac{1}{Re} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \mathbf{D} \mathbf{U} \quad (9)$$

$$\mathbf{U} = \begin{bmatrix} p \\ u \\ v \end{bmatrix}, \quad \mathbf{E} = \begin{bmatrix} u/\beta \\ p + u^2 \\ uv \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} v/\beta \\ uv \\ p + v^2 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (10)$$

This system can now be solved iteratively using a compressible flow solver, where the artificial time derivative of pressure is updated until convergence is achieved, yielding the steady-state solution for the incompressible flow.

2. Derivation and Discretization of Governing Equations

In order to implement the governing equations, the equations need to first be discretized with respect to space and time. In this implementation, central differences on a collocated mesh are used to discretize the governing equations in the domain. Then, the explicit Euler method is used to advance the solution U after every time step.

We discretize the height of the channel $H = 1$ into $N_y = 20$ equally spaced cells, yielding a vertical grid spacing $\Delta y = \frac{H}{N_y} = 0.05$. The length of the channel L is discretized into N_x equally spaced cells, yielding a horizontal grid spacing $\Delta x = \frac{L}{N_x}$. We restrict ourselves to using square cells so that $\Delta x = \Delta y$. The first and second derivatives can now be approximated on this discretized domain using second-order central differences:

$$\frac{\partial}{\partial x} \approx \frac{\delta}{\delta x} = \frac{\square_{i+1,j} - \square_{i-1,j}}{2\Delta x} \quad \frac{\partial^2}{\partial x^2} \approx \frac{\delta^2}{\delta x^2} = \frac{\square_{i+1,j} - 2\square_{i,j} + \square_{i-1,j}}{\Delta x^2} \quad (11)$$

$$\frac{\partial}{\partial y} \approx \frac{\delta}{\delta y} = \frac{\square_{i,j+1} - \square_{i,j-1}}{2\Delta y} \quad \frac{\partial^2}{\partial y^2} \approx \frac{\delta^2}{\delta y^2} = \frac{\square_{i,j+1} - 2\square_{i,j} + \square_{i,j-1}}{\Delta y^2} \quad (12)$$

Depending on the context, $\square_{i,j}$ denotes the numerical approximation of (the elements of) $\mathbf{E}$, $\mathbf{F}$, or $\mathbf{DU}$ at spatial point (i, j) . In a similar way, the artificial time derivative can be discretized:

$$\frac{\partial \mathbf{U}}{\partial \tilde{t}} \approx \frac{\mathbf{U}^{n+1} - \mathbf{U}^n}{\Delta \tilde{t}} \quad (13)$$

Here, $\mathbf{U}^n$ denotes the numerical approximation of (the elements of) $\mathbf{U}$ at time point n and $\Delta \tilde{t}$ denote the size of this fictitious time step. Equations (9) and (13) can be combined to solve for $\mathbf{U}^{n+1}$. If we take the right hand side (RHS) of (14) to be evaluated at time point n , we obtain the explicit Euler scheme.

$$\mathbf{U}^{n+1} = \mathbf{U}^n + \Delta \tilde{t} \left\{ -\frac{\partial \mathbf{E}}{\partial x} - \frac{\partial \mathbf{F}}{\partial y} + \frac{1}{Re} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \mathbf{DU} \right\}^n \quad (14)$$

After updating $\mathbf{U}^{n+1}$, we modify $\mathbf{U}^{n+1}$ to have the proper boundary conditions. Since we are using a collocated mesh, we must use a layer of ghost nodes all around the domain to achieve second-order spatial accuracy. In this way, the domain boundaries are located at the interface between the ghost cell and the outermost interior cell. The inlet boundary conditions are discretized as:

$$\frac{p_{\text{ghost}} - p_{\text{interior}}}{\Delta x} = 0 \iff p_{\text{ghost}} = p_{\text{interior}} \quad (15)$$

$$\frac{u_{\text{ghost}} + u_{\text{interior}}}{2} = 0 \iff u_{\text{ghost}} = -u_{\text{interior}} \quad (16)$$

$$\frac{v_{\text{ghost}} + v_{\text{interior}}}{2} = 0 \iff v_{\text{ghost}} = -v_{\text{interior}} \quad (17)$$

The outlet boundary conditions are discretized as:

$$\frac{p_{\text{ghost}} + p_{\text{interior}}}{2} = 0 \iff p_{\text{ghost}} = -p_{\text{interior}} \quad (18)$$

$$\frac{u_{\text{ghost}} - u_{\text{interior}}}{\Delta x} = 0 \iff u_{\text{ghost}} = u_{\text{interior}} \quad (19)$$

$$\frac{v_{\text{ghost}} - v_{\text{interior}}}{\Delta x} = 0 \iff v_{\text{ghost}} = v_{\text{interior}} \quad (20)$$

The no-slip wall boundary conditions are discretized as:

$$\frac{p_{\text{ghost}} - p_{\text{interior}}}{\Delta y} = 0 \iff p_{\text{ghost}} = p_{\text{interior}} \quad (21)$$

$$\frac{u_{\text{ghost}} + u_{\text{interior}}}{2} = 0 \iff u_{\text{ghost}} = -u_{\text{interior}} \quad (22)$$

$$\frac{v_{\text{ghost}} + v_{\text{interior}}}{2} = 0 \iff v_{\text{ghost}} = -v_{\text{interior}} \quad (23)$$

Discretizing the boundary conditions is also necessary since the solution values at the ghost cells cannot be prescribed by the central difference approximations. Another issue that arises from using a collocated mesh is odd-even decoupling, causing high-frequency oscillations in the pressure field. To dampen these oscillations, we add a smoothing term to the pressure field:

$$p = p^* - \varepsilon_e \left(\frac{\partial^4}{\partial x^4} + \frac{\partial^4}{\partial y^4} \right) p^* \quad (24)$$

where ε_e is another user-defined coefficient. The fourth-order derivatives are discretized as follows:

$$\frac{\partial^4 p}{\partial x^4} \approx p_{i+2,j} - 4p_{i+1,j} + 6p_{i,j} - 4p_{i-1,j} + p_{i-2,j} \quad (25)$$

$$\frac{\partial^4 p}{\partial y^4} \approx p_{i,j+2} - 4p_{i,j+1} + 6p_{i,j} - 4p_{i,j-1} + p_{i,j-2} \quad (26)$$

Unlike other implementations, this code absorbs the Δx^4 and Δy^4 factors into ε_e to prevent the derivative from growing too large especially for small Δx and Δy . Putting it all together, we obtain an algorithm which can update the elements of $\mathbf{U}$ after every time step.

1. $\mathbf{U}^{n+1} = \mathbf{U}^n + \Delta t \left\{ -\frac{\delta \mathbf{E}}{\delta x} - \frac{\delta \mathbf{F}}{\delta y} + \frac{1}{Re} \left(\frac{\delta^2 \mathbf{D}\mathbf{U}}{\delta x^2} + \frac{\delta^2 \mathbf{D}\mathbf{U}}{\delta y^2} \right) \right\}^n$ (Updates interior cell values.)
2. Add smoothing term to pressure field. (Dampen high-frequency oscillations.)
3. Reinforce boundary conditions. (Updates ghost cell values.)

3. Determining the Channel Length of Fully Developed Flow

The dimensionless analytical solution for the velocity profile of fully developed 2-D steady-state channel flow is given by the parabolic profile:

$$u(y) = 6y(1 - y), \quad v(y) = 0, \quad y \in [0, 1] \quad (27)$$

By varying the length of the channel L and comparing the numerical solution against (27), we can determine at what length the flow becomes fully developed. In this test, Re is set to 20 and the mean absolute error (MAE) between the numerical solution at $x = L$ and the analytical solution is computed and plotted for $L = 0.5, 1.0, 1.5, 2.0, 2.5$.

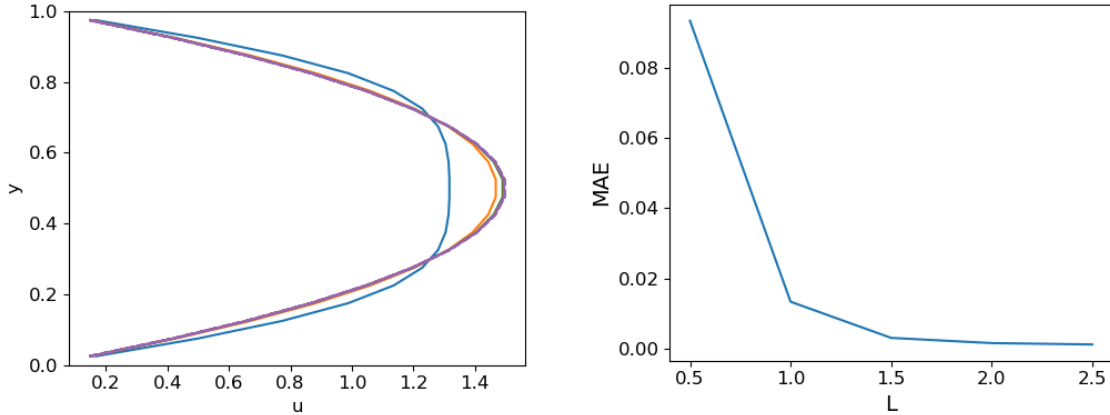


Figure 1: **Left:** Numerical velocity profiles at $x = L$ given $Re = 20$ and $L = 0.5$ (solid blue), 1.0 (solid orange), 1.5 (solid green), 2.0 (solid red), 2.5 (solid purple). Although hard to see, the analytical solution is given by the dotted purple line. **Right:** MAE between analytical solution and numerical solution at $x = L$.

From the left graph of **Figure 1**, the analytical solution and the numerical solution at $x \approx 2.5$ overlap significantly, suggesting that the flow becomes fully developed when $L \approx 2.5$. This is also indicated by exceptionally low MAE of around 10^{-3} in the right graph.

4. Comparison of Numerical and Analytical Velocity and Pressure Profiles

In this next test, the numerical velocity profiles were plotted at $x = 2.5, 1.25, 0.625, 0.3125, 0.15625$ in a channel with $Re = 20, L = 2.5$. Additionally, the numerical pressure profile along the channel's axis of symmetry $y = 0.5$ is plotted.

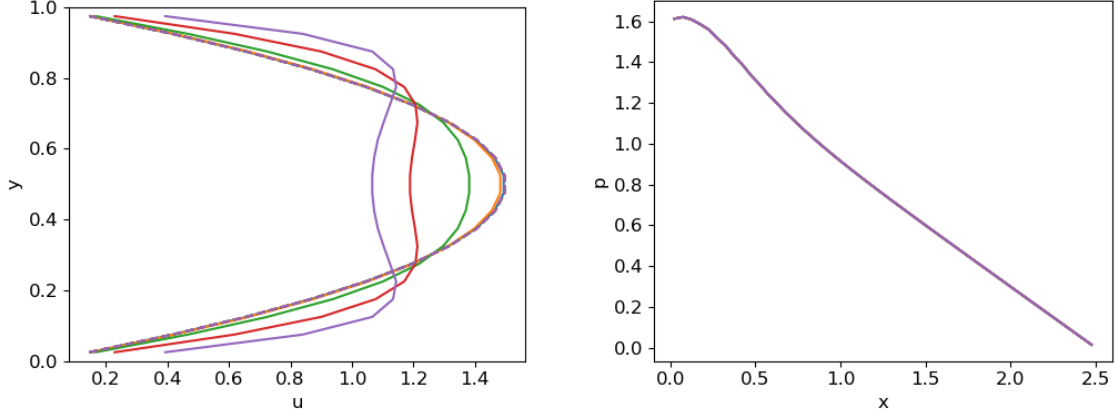


Figure 2: **Left:** Numerical velocity profiles for $Re = 20, L = 2.5$ at $x = 2.5$ (solid blue), 1.25 (solid orange), 0.625 (solid green), 0.3125 (solid red), 0.15625 (solid purple). Although hard to see, the analytical solution for fully developed flow is given by the dotted purple line. **Right:** Numerical pressure profile for $Re = 20, L = 2.5$ at $y = 0.5$.

The left graph of **Figure 2** shows that as we move from the inlet to the outlet, flow becomes more developed, which is also expected in real channel flow. This is because the viscosity of the fluid causes the fluid near the wall to slow down; this difference in fluid velocity at the center and at the wall gets larger as the fluid continues to flow down the channel.

The right graph of **Figure 2** shows that the pressure decreases from some finite value at $x = 0$ to 0 at $x = 2.5$ at a roughly linear rate, which is also expected in real channel flow. Fully developed channel flow is driven by a constant pressure gradient, which is accurately reflected in this simulation. Near $x = 0$, there is a nonphysical bump in the pressure profile. This is caused by the nonphysical inlet boundary condition $\frac{\partial p}{\partial x} = 0$ used to partially reduce the high-frequency oscillations that can develop near the inlet.

5. Stability Criteria and Time Step Size

Explicit schemes like the one implemented here are stable only when the Mach number $Ma = |u|/\tilde{a} < 1$ and the CFL number $CFL = \tilde{a}\Delta\tilde{t}/\Delta x < 1$, where $\tilde{a} = 1/\sqrt{\beta}$ is the artificial speed of sound. Because of these restrictions, the artificial time step size $\Delta\tilde{t}$ must be chosen carefully to prevent the solution from diverging.

We know $\Delta x = \Delta y = 0.05$ and $u_{\max} = 1.5$ from the analytical solution; thus, we can easily determine $\tilde{a}, \beta, \Delta\tilde{t}$ to keep the explicit solver stable. Since $|u| < u_{\max}$ for stable solutions, we choose $\tilde{a} = 2$ (or $\beta = 0.25$) to keep the Mach number safely below 1. From this, we know that the time step size must be $\Delta\tilde{t} < \Delta x/\tilde{a} = 0.05/2 = 0.025$. The simulations throughout this report use $\Delta\tilde{t} = 0.001$, satisfying the CFL number criterion.

6. Investigation of the Spatial Convergency

To evaluate the spatial convergency of the solver, MAE between the analytical solution and the numerical solution at $x = L$ for $N = 10, 20, 40, 60$ given $Re = 20, L = 2.5$. In **Figure 3**, it can be seen that the explicit scheme is roughly second-order accurate with respect to Δx since the relationship between MAE and Δx can be approximated by a power relationship with exponent 2.2.

This observation also matches the theoretical prediction of the spatial convergency. Since the finite difference approximations of the spatial derivatives and the boundary conditions are second-order accurate with respect to Δx , we expect the explicit scheme to be second-order accurate overall.

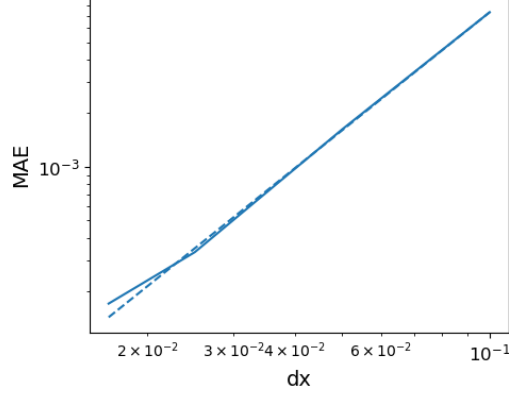


Figure 3: Log-log plot of MAE between analytical solution and numerical solution at $x = L$. For reference, the dotted line is a power law relationship with an exponent of 2.2

7. Evaluation of the Entrance Length

As fluid flows through the channel, the no-slip condition at the walls creates a boundary layer where the fluid is slowed down by friction. This boundary layer grows thicker as the fluid moves downstream until the layers from the top and bottom meet in the middle, creating the final parabolic velocity profile. When Re increases and inertial forces begin to dominate over viscous forces, it takes much longer for the viscous friction at the walls to penetrate into the core and reshape the profile. As a result, the fluid must travel a greater distance down the channel before it becomes fully developed, increasing L_e according to the empirical law:

$$L_e = [0.631^{1.6} + (0.0442Re)^{1.6}]^{\frac{1}{1.6}} \quad (28)$$

To verify this phenomenon, the simulation was ran with $L = 20$ and $Re = 50, 100, 200, 300, 400, 500$. To determine L_e we look for the point at which the velocity profile stops changing in the x -direction i.e. $\frac{\partial u}{\partial x} < 0.01$.

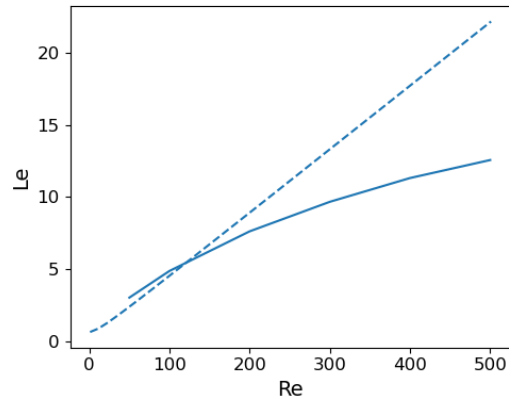


Figure 4: **Solid Line:** Numerical results of increasing L_e with respect to Re . **Dotted Line:** Empirical law determined by (28).

It is clear from **Figure 4** that the numerical results diverge significantly from the empirical law. Although the exact reason as to why is unknown, one possible explanation is that in between these runs, the smoothing factor ε_e was increased to allow the simulations with $Re \geq 300$ to converge. (More specifically, $\varepsilon_e = 10^{-4}$ for $Re < 300$, $\varepsilon_e = 10^{-3}$ for $Re = 300$, and $\varepsilon_e = 10^{-2}$ for $Re > 300$.) While it dampens oscillations, it also artificially diffuses momentum. This can make the flow appear more developed than it actually is, effectively shortening the calculated entrance length.